

the two groups, which is consistent with the hypothesis that radiologists participating in the study were of similar skill and exerted similar effort as the radiologists completing the original reads.

4.2 How do Radiologists Respond to AI and Contextual Information?

We now describe the effects of our information treatments estimated using the following specification:

$$Y_{irt} = \gamma_{h_i} + \gamma_{CH} \cdot d_{CH}(t) + \gamma_{AI} \cdot d_{AI}(t) + \gamma_{AI \times CH} \cdot d_{CH}(t) \cdot d_{AI}(t) + \varepsilon_{irt},$$

where Y_{rit} is an outcome variable of interest for radiologist r diagnosing patient case-pathology i and treatment t , and γ_{h_i} are pathology fixed effects since there are multiple pathologies h_i for each case in this pooled analysis. Treatments t vary by whether or not clinical history is provided $d_{CH}(t) \in \{0, 1\}$ and whether or not AI information is provided $d_{AI}(t) \in \{0, 1\}$. We report two-way clustered standard errors at the radiologist and patient-case level. The specification omits radiologist-specific fixed effects because the treatments are balanced within radiologists. Cases are also balanced across treatments (see appendix C.1), which suggests that case randomization was successful. We will also compute conditional treatment effects given ranges of the AI signal s_i^A that are grouped based on $\pi(\omega_i = 1 | s_i^A)$.

Table 2: Average treatment effects

Treatment	Deviation from AI		Deviation from Ground Truth		Effort Measures			
	All Designs	Design 1	All Designs	Design 1	All Designs		Design 1	
					Active Time	Clicks	Active Time	Clicks
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
AI $\times$ CH	0.001 (0.003)	−0.001 (0.006)	0.004 (0.005)	0.003 (0.012)	−0.86 (3.68)	0.22 (0.82)	0.04 (6.94)	0.32 (1.63)
AI	−0.038 (0.004)	−0.036 (0.006)	0.002 (0.004)	0.002 (0.008)	6.75 (2.47)	1.34 (0.56)	6.26 (4.71)	1.53 (1.05)
CH	−0.001 (0.002)	−0.004 (0.004)	−0.011 (0.004)	−0.015 (0.008)	7.37 (2.53)	0.16 (0.55)	6.53 (4.78)	0.09 (1.08)
CONTROL MEAN	0.211 (0.007)	0.221 (0.008)	0.227 (0.010)	0.224 (0.011)	157.57 (4.46)	44.25 (1.21)	161.56 (7.05)	40.71 (1.53)
PATHOLOGY FE	Yes	Yes	Yes	Yes	-	-	-	-
OBSERVATIONS	36280	13440	36280	13440	14635	14635	6718	6718

Note: This table summarizes the average treatment effects (ATE) of different information environments on the absolute value of the difference between the radiologist probability and AI probability (column (1) and (2)); absolute value of the difference between the radiologist probability and the ground truth (columns (3) and (4)); and radiologists' effort measured in terms of active time and clicks (columns (5), (6), (7) and (8)). We either pool across all designs (All Designs) or condition on only design 1. Results on effort measure excludes five patient-cases with unaccounted time measure, and observations from design 3 because of learning effects in this set-up. The results are for the two top-level pathologies, airspace opacity and cardiomediastinal abnormality. Standard errors are two-way clustered at the radiologist and patient-case level in parenthesis. Robustness by design can be found in section C.4.1.

The above analysis focuses on the effects on the marginal distributions of the outcome variables Y_{irt} for each pathology. Thus, the specification abstracts away from interactions between pathologies in the effects on information provision, for example due to potential dependence between the predictions and decisions. In section 5, we will present evidence showing that the best fitting model has radiologists updating their beliefs as-if they do not account for dependence between pathologies.

The treatment effect analysis in the main text pools the three experimental designs and does not condition on the sequence in which subjects encounter information treatments. Appendix C.4.5 shows that our results are robust to including controls for order effects. The estimates from all three designs are similar to each other. They are also statistically indistinguishable from those that use only an across participant comparison from the first treatment encountered in design 1 (appendix C.4.3).

4.2.1 Do Radiologists Utilize AI Predictions?

We begin by testing whether radiologists respond to the information that the AI provides. The left panel in table 2 shows how the different information environments affect the disagreement of the radiologists' report with the AI's assessment, which is defined as $Y_{irt} = |p_r(\omega_i = 1 | s_{irt}) - \pi(\omega_i = 1 | s_i^A)|$. The term $p_r(\omega_i = 1 | s_{irt})$ is elicited whereas $\pi(\omega_i = 1 | s_i^A)$ is the AI's predicted probability that $\omega_i = 1$. When t indicates that AI assistance is provided, then $s_{irt} = (s_{ir}^E, s_i^A)$, and when t indicates that AI assistance is not provided, then $s_{irt} = s_{ir}^E$. The signal s^E also depends on whether contextual information is provided.

The results show that radiologists utilize AI assistance: their predictions are closer to the AI predictions when provided with assistance. To see this, observe that the control means for the deviation from the AI are approximately 0.21 for both when we pool designs and for design 1 only. Treatments where AI is provided reduce this baseline average deviation by 18% (all designs) and 16% (design 1). ($p < 0.01$).

4.2.2 Treatment Effects on Diagnostic Performance

We next ask whether the information treatments affect radiologists' diagnostic performance. As a measure, we consider the absolute deviation of the radiologist's probability report from the binary ground truth, $Y_{irt} = |p_r(\omega_i = 1 | s_{irt}) - \omega_i|$ where lower values implies better performance. Appendix C.4.2 contains results that use a continuous ground truth, which are

qualitatively similar. The middle panel of table 2 shows the average treatment effects on performance.

Our results indicate that while access to contextual information improves performance on average, AI assistance does not. We find that access to clinical history reduces the deviation from the ground truth by 4.8% ($p < 0.05$) of the control mean if we pool designs and by 6.7% in design 1 ($p < 0.05$). In contrast, the effects for AI are close to zero and not statistically significant. In light of the findings in the previous two sections — that the AI is more accurate than most radiologists and that radiologists move their assessments toward the AI—it is puzzling that the AI information does not improve accuracy.

This contradiction occurs because the average treatment effects mask significant heterogeneity in treatment effects. Our within-participant designs—designs 2 and 3—allow us to estimate conditional treatment effects given radiologists’ predictions without AI assistance. Specifically, we partition cases based on the expert’s signal into five equally spaced bins of $\pi_r(\omega_i = 1 | s_{ir}^E)$. Figure 2 shows the conditional treatment effects of providing AI assistance on diagnostic performance. Panel (a) shows the deviation from the ground truth and panel (b) shows the probability of incorrect decision. We find that providing AI assistance in cases when the radiologist is uncertain improves performance on both metrics, whereas AI assistance is harmful when the radiologist is close to certain that the pathology is not present for a given case. Given an average reported probability of 25% or less, the vast majority of cases fall in the first bin yielding a small average treatment effect that masks important heterogeneity. The result that AI assistance can decrease performance rejects a model in which radiologists are Bayesians with correct beliefs.